WORKED SOLUTION

# STEP I 1990, Question 7

Pure • First Order Linear ODEs

**STEP I 1990, Question 7**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q7

## The key idea

Writing $y = uv$ gives us two unknown functions where we only need one, so we can afford to impose an extra condition on one of them. The natural choice is to make $u$ kill the term containing $P$, which leaves a differential equation that integrates directly. This is the integrating-factor method in disguise: $1/u$ is exactly the integrating factor.

## Solution

### The substitution

With $y = uv$, the product rule gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u\dfrac{\mathrm{d}v}{\mathrm{d}x} + v\dfrac{\mathrm{d}u}{\mathrm{d}x}$, so the equation becomes

$$u\frac{\mathrm{d}v}{\mathrm{d}x} + v\left(\frac{\mathrm{d}u}{\mathrm{d}x} + Pu\right) = Q.$$

If we choose $u$ to satisfy the first order equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} + Pu = 0,$$

the bracket vanishes and we are left with

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{Q}{u},$$

an equation for $\frac{\mathrm{d}v}{\mathrm{d}x}$ in terms of $x$, $Q$ and $u$ only, as required. (We need $u \neq 0$ to divide; the non-zero solutions of $u' + Pu = 0$ are $u = A\mathrm{e}^{-\int P\,\mathrm{d}x}$ with $A \neq 0$, which never vanish.)

### The particular equation

Here $P = -\dfrac{2}{x+1}$ and $Q = (x+1)^{5/2}$. Note that $Q$ is only defined (as a real number) for $x \geq -1$ and $P$ is undefined at $x = -1$, so we work on $x > -1$, where $x + 1 > 0$.

**Ask yourself** Do we need the general solution for $u$, or will any one solution do?

Any one non-zero solution will do: the arbitrary constant in the final answer appears when we integrate for $v$. So we solve

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u}{x+1}$$

by separating variables: $\ln|u| = 2\ln(x+1) + \text{const}$, and we take

$$u = (x+1)^2.$$

(Check: $u' = 2(x+1) = \frac{2u}{x+1}$.) Then

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{(x+1)^{5/2}}{(x+1)^2} = (x+1)^{1/2}, \qquad \text{so} \qquad v = \tfrac{2}{3}(x+1)^{3/2} + C.$$

Hence the general solution is

$$y = uv = \tfrac{2}{3}(x+1)^{7/2} + C(x+1)^2.$$

Applying $y(1) = 0$:

$$0 = \tfrac{2}{3}\cdot 2^{7/2} + 4C \quad \Longrightarrow \quad C = -\tfrac{1}{6}\cdot 2^{7/2} = -\tfrac{1}{6}\cdot 8\sqrt{2} = -\frac{4\sqrt{2}}{3}.$$

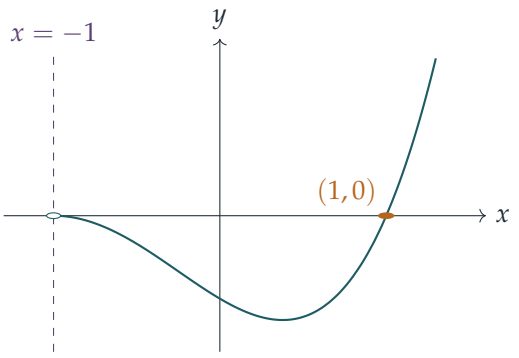
Therefore

$$\boxed{y = \tfrac{2}{3}(x+1)^{7/2} - \tfrac{4\sqrt{2}}{3}(x+1)^2 = \tfrac{2}{3}(x+1)^2\left[(x+1)^{3/2} - 2\sqrt{2}\right]}$$

#### Where the solution is valid

**Ask yourself** The formula for $y$ makes sense at $x = -1$. Why isn't $x = -1$ included?

The solution is valid for $x > -1$. For $x < -1$ the right-hand side $(x+1)^{5/2}$ is not real, so the differential equation itself means nothing there (and nor does our $(x+1)^{7/2}$). At $x = -1$ the term $\frac{2y}{x+1}$ is undefined, so the equation doesn't hold there either, even though our formula gives $y \to 0$ as $x \to -1^+$. On the whole interval $x > -1$, with $x + 1 > 0$, every step above is legitimate, and the solution passes through the given point $x = 1$, which lies in this interval.




The open circle marks the excluded endpoint $x = -1$.

## Checks

- Substitute back: $y' = \frac{7}{3}(x+1)^{5/2} - \frac{8\sqrt{2}}{3}(x+1)$ and $\frac{2y}{x+1} = \frac{4}{3}(x+1)^{5/2} - \frac{8\sqrt{2}}{3}(x+1)$; their difference is $(x+1)^{5/2}$, as required.
- $y(1) = \frac{2}{3} \cdot 4\left(2\sqrt{2} - 2\sqrt{2}\right) = 0$.
- Special values: $y(0) = \frac{2-4\sqrt{2}}{3} \approx -1.219$ and $y(3) = \frac{256-64\sqrt{2}}{3} \approx 55.16$. A numerical (Runge–Kutta) integration of the equation from $y(1) = 0$ agrees with both to nine figures.

## Common slips

- Getting the sign of $P$ wrong: here $P = -\frac{2}{x+1}$, so $u = (x+1)^2$, not $(x+1)^{-2}$.
- Forgetting the constant when integrating for $v$, which leaves no freedom to fit $y(1) = 0$.
- Giving $x \geq -1$ as the range: the equation is undefined at $x = -1$.

## Beyond STEP

Choosing $u$ to remove a troublesome term is the idea behind *variation of parameters*: solve the homogeneous equation first, then let the "constant" vary to absorb the forcing. The same trick works for second order linear equations, and the question of where a solution is valid grows into the existence and uniqueness theorem, which guarantees a unique solution on any interval where $P$ and $Q$ are continuous. Boyce and DiPrima, *Elementary Differential Equations*, covers both.

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